



Options Markets

Review of Option Types

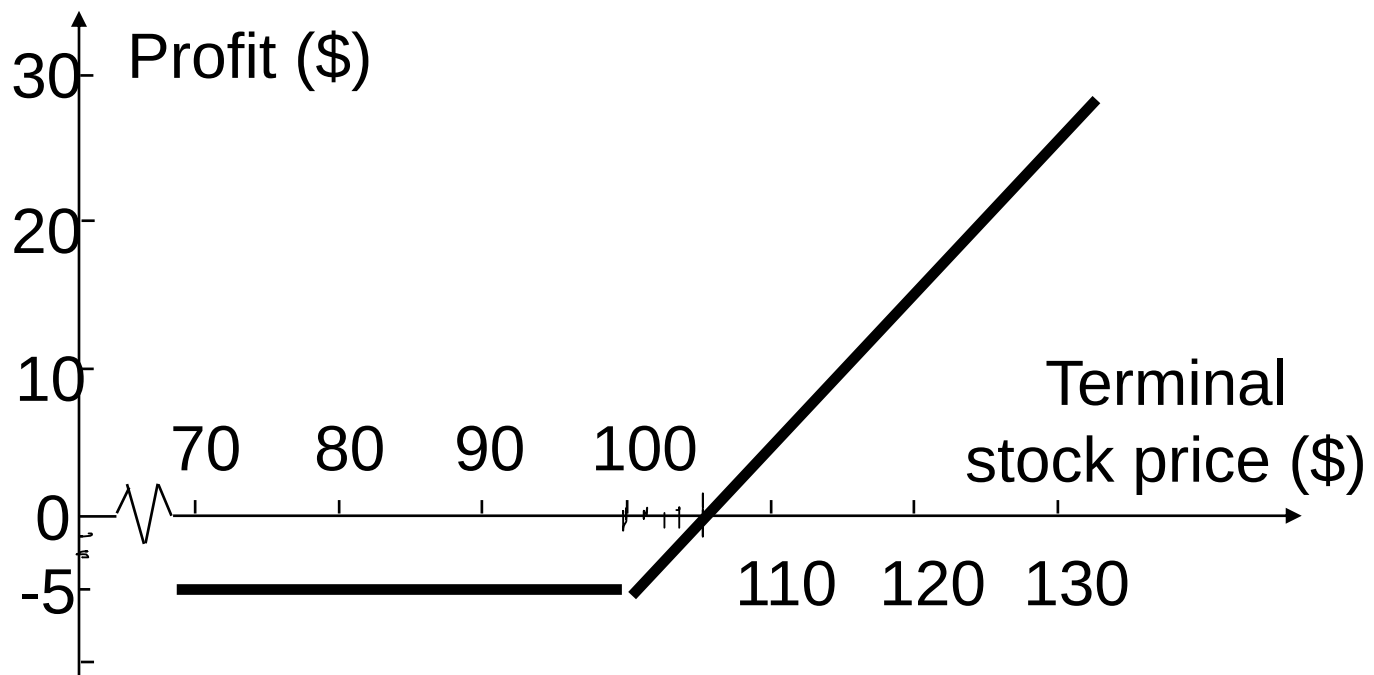
- A call is an option to buy
- A put is an option to sell
- A European option can be exercised only at the end of its life
- An American option can be exercised at any time

Option Positions

- Long call
- Long put
- Short call
- Short put

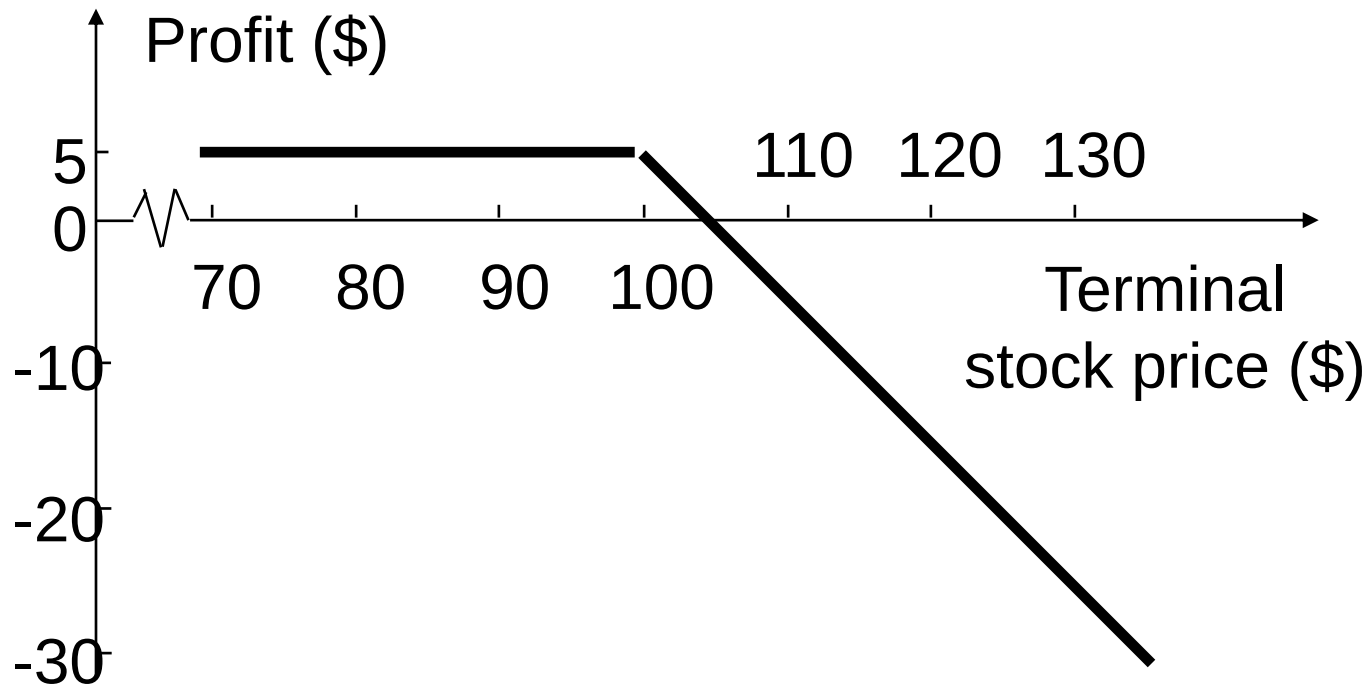
Long Call

Profit from buying one European call option: option price = \$5, strike price = \$100, option life = 2 months



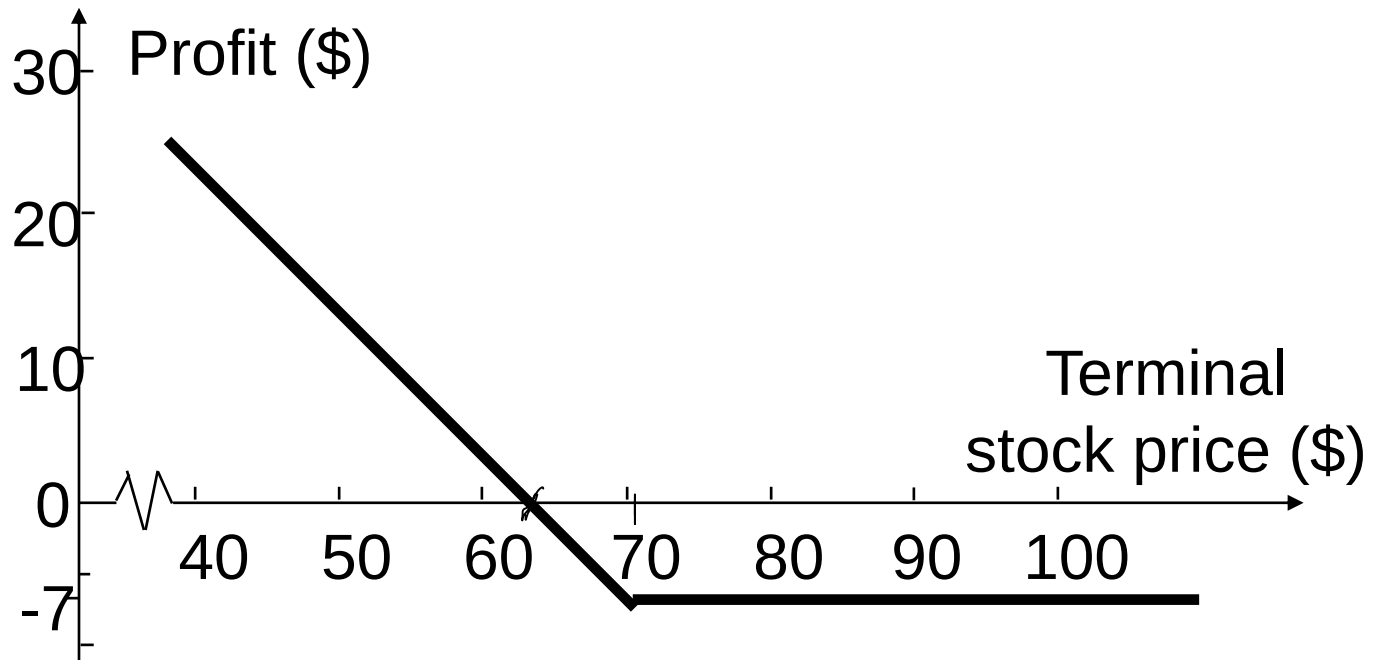
Short Call

Profit from writing one European call option: option price = \$5, strike price = \$100



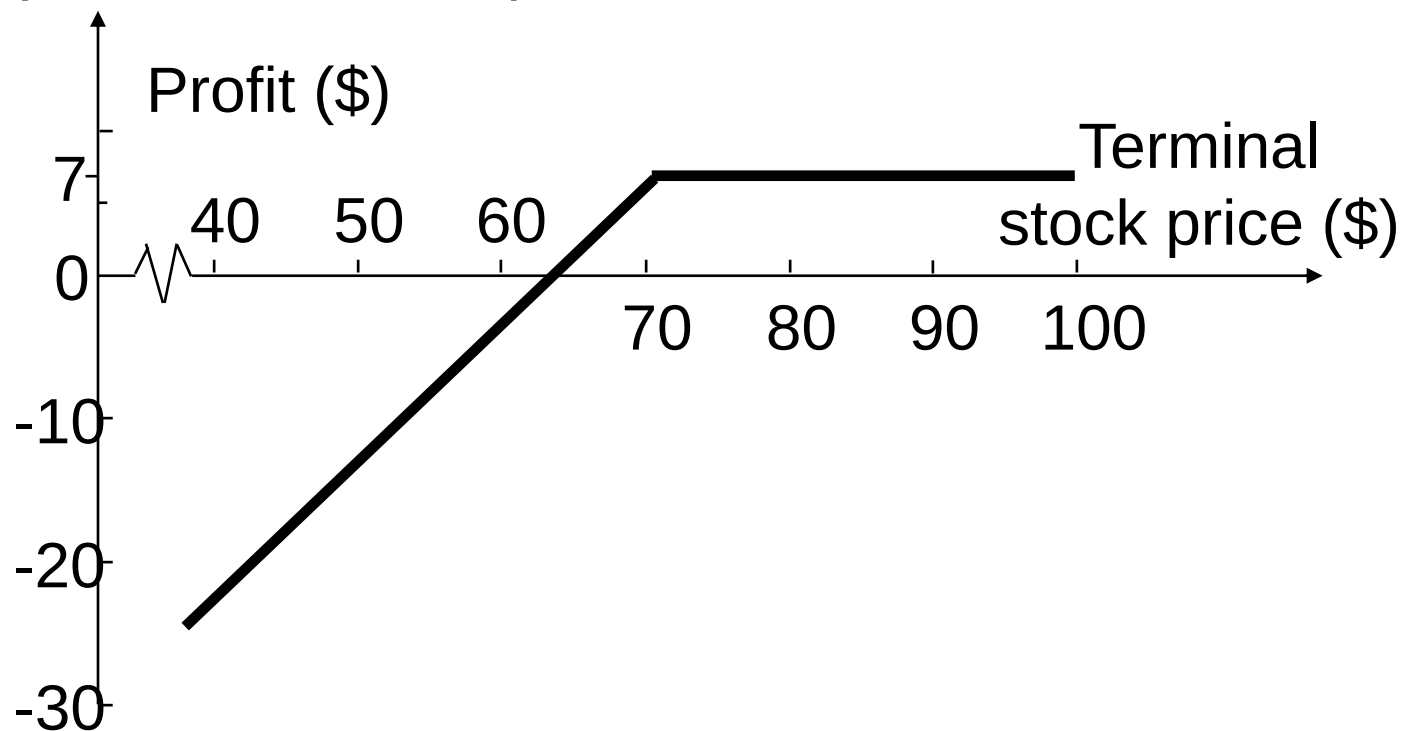
Long Put

Profit from buying a European put option: option price = \$7, strike price = \$70



Short Put

Profit from writing a European put option: option price = \$7, strike price = \$70

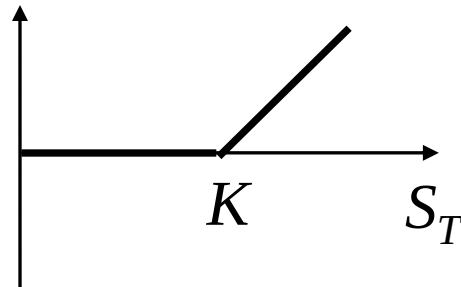


Payoffs from Options

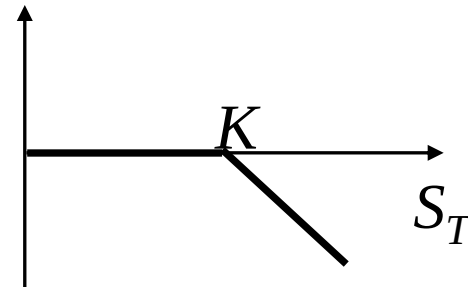
What is the Option Position in Each Case?

K = Strike price, S_T = Price of asset at maturity

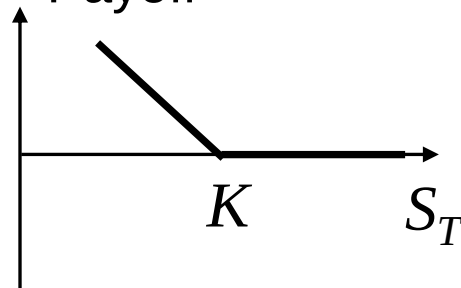
Payoff



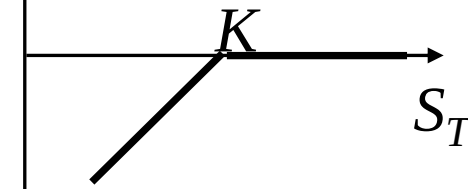
Payoff



Payoff



Payoff





Specification of Exchange-Traded Options

- Expiration date
- Strike price
- European or American
- Call or Put (option class)

Terminology

Moneyiness :

- At-the-money option
- In-the-money option
- Out-of-the-money option

Terminology

(continued)

- Option class
- Option series
- Intrinsic value
- Time value

Intrinsic value & Time value

- The intrinsic value of an option represents the current value of the option, or in other words how much in the money it is. When an option is in the money, this means that it has a positive payoff for the buyer.
- In the money call options: $\text{Intrinsic Value} = \text{Price of Underlying Asset} - \text{Strike Price}$
- In the money put options: $\text{Intrinsic Value} = \text{Strike Price} - \text{Price of Underlying Asset}$

Time value

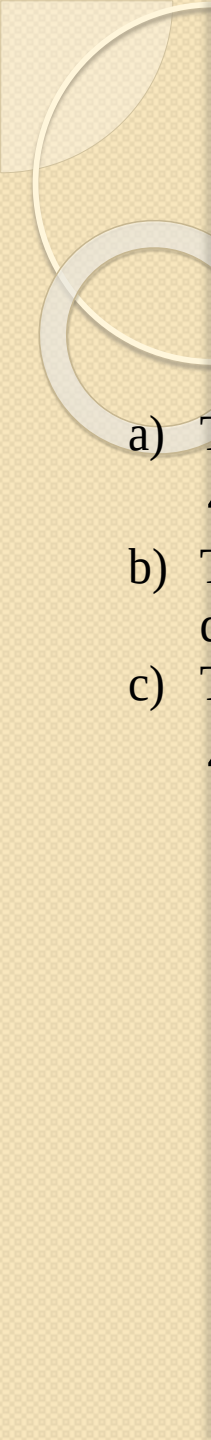
- The time value of an option is an additional amount an investor is willing to pay over the current intrinsic value. Investors are willing to pay this because an option could increase in value before its expiration date.
- $\text{Time Value} = \text{Option Premium} - \text{Intrinsic Value}$
- $\text{Option Premium} = \text{Intrinsic Value} + \text{Time Value}$

Dividends & Stock Splits

- Suppose you own N options with a strike price of K :
 - No adjustments are made to the option terms for cash dividends
 - When there is an n -for- m stock split,
 - the strike price is reduced to mK/n
 - the no. of options is increased to nN/m
 - Stock dividends are handled similarly to stock splits

Dividends & Stock Splits

- *Consider an exchange-traded call option contract to buy 500 shares with a strike price of \$40 and maturity in four months. Explain how the terms of the option contract change when there is*
 - a) A 10% stock dividend*
 - b) A 10% cash dividend*
 - c) A 4-for-1 stock split*

- 
- a) The option contract becomes one to buy $500 \times 1.1 = 550$ shares with an exercise price $40/1.1 = 36.36$.
 - b) There is no effect. The terms of an options contract are not normally adjusted for cash dividends.
 - c) The option contract becomes one to buy $500 \times 4 = 2,000$ shares with an exercise price of $40/4 = \$10$.

Market Makers

- Most exchanges use market makers to facilitate options trading
- A market maker quotes both bid and ask prices when requested
- The market maker does not know whether the individual requesting the quotes wants to buy or sell

Notation

c :	European call option price
p :	European put option price
S_0 :	Stock price today
K :	Strike price
T :	Life of option
σ :	Volatility of stock price

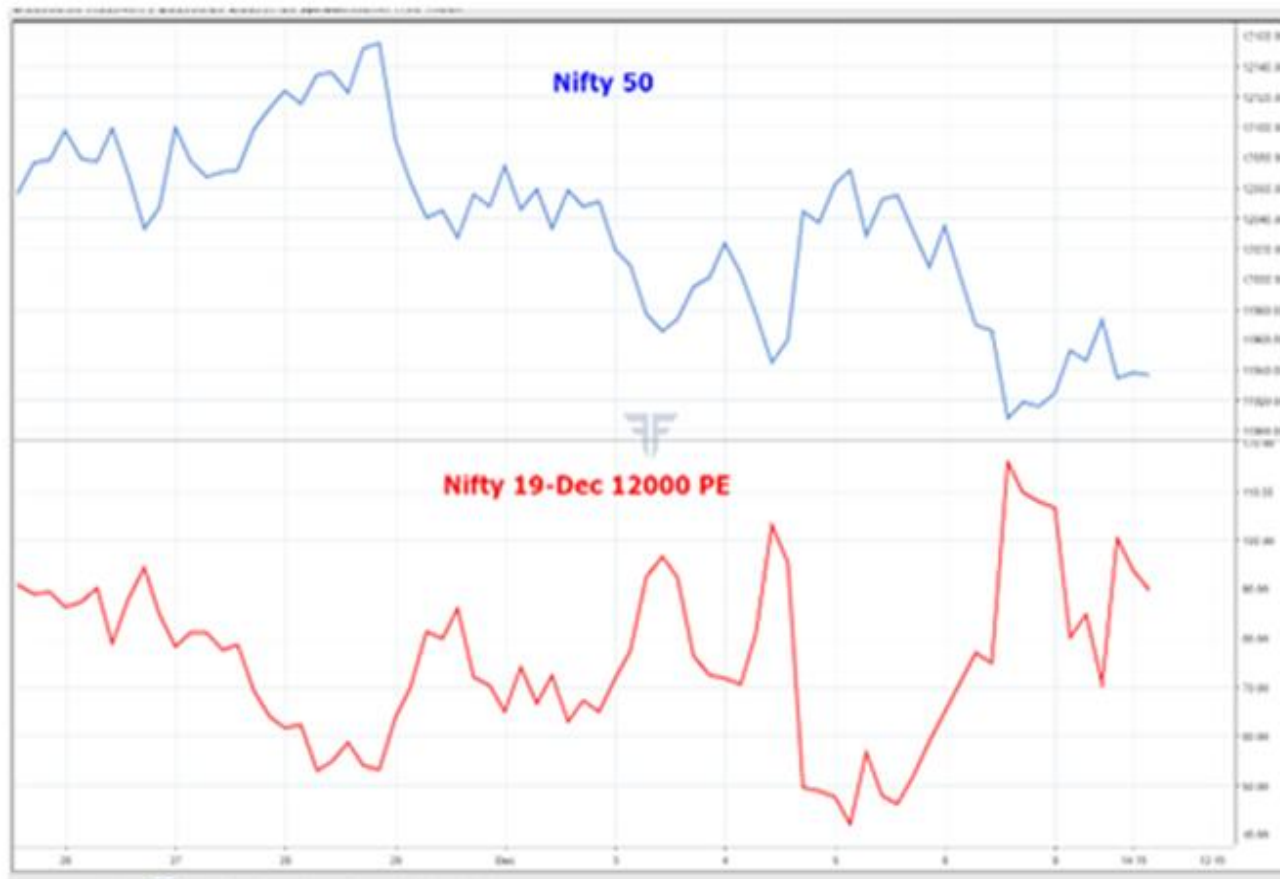
C :	American call option price
P :	American put option price
S_T :	Stock price at option maturity
D :	PV of dividends paid during life of option
r :	Risk-free rate for maturity T with cont. comp.

Effect of Variables on Option Pricing

Variable	c	p	C	P
S_0	+	-	+	-
K	-	+	-	+
T	?	?	+	+
σ	+	+	+	+
r	+	-	+	-
D	-	+	-	+



The chart above compares intraday movement in Nifty index with a corresponding movement in the Nifty 12000 Call Option (denoted as CE). Observe in the chart that as the price of Nifty (blue line) declined, so did the price of the Nifty 12000 CE (red line). Also notice the minor upticks in the option price whenever Nifty recovered from lows.

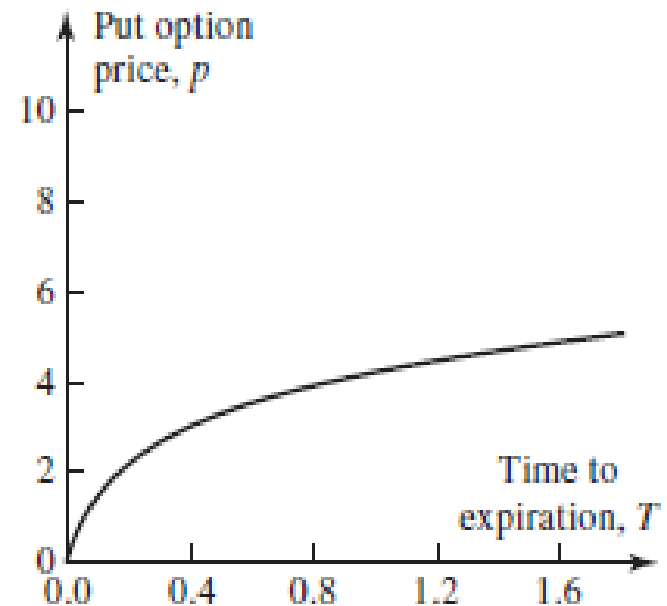
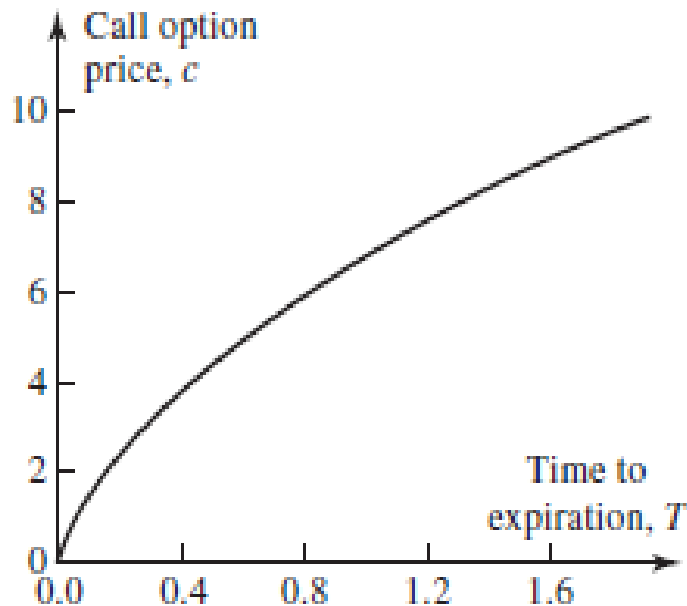


Meanwhile, the chart above compares intraday movements in Nifty index with a corresponding movement in the Nifty 12000 Put Option (denoted as PE). Notice that as the price of Nifty dropped (blue line), the price of the Nifty 12000 PE went higher (red line). Also observe that intermittent recoveries in Nifty were accompanied by a noticeable decline in option price.

CALLS											PUTS											
	OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID PRICE	ASK PRICE	ASK QTY	STRIKE PRICE	BID QTY	BID PRICE	ASK PRICE	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI	
	439	27	153	-	468.55	-57.15	100	455.35	490.25	50	17,550.00	50	1.50	2.10	700	-2.80	1.55	24.38	1,37,673	1,049	13,275	
	891	-126	5,750	-	408.90	-56.20	50	407.35	410.90	100	17,600.00	1,650	2.20	2.40	1,350	-2.85	2.40	23.74	4,71,708	-4,806	41,920	
	560	-55	1,097	-	359.25	-53.40	100	357.50	360.10	50	17,650.00	200	2.95	3.05	500	-3.65	2.95	22.15	3,04,251	-81	22,487	
	6,544	-521	19,672	-	310.00	-58.75	50	310.25	312.00	1,800	17,700.00	1,600	4.40	4.50	200	-4.50	4.40	21.26	7,25,697	5,250	59,023	
	1,553	620	16,396	-	262.70	-62.80	1,000	260.20	266.05	200	17,750.00	150	6.40	7.00	300	-5.35	7.00	20.67	4,87,773	11,330	32,548	
	15,273	366	1,42,281	-	217.70	-56.95	50	216.95	217.90	300	17,800.00	250	10.35	10.70	1,400	-6.95	10.35	19.73	12,16,068	12,370	76,303	
	4,448	-1,292	92,987	9.66	173.10	-59.90	100	173.10	174.55	100	17,850.00	50	16.75	16.95	200	-6.50	16.95	19.45	6,96,109	6,456	35,086	
	20,777	-12,029	7,61,899	13.48	133.40	-56.90	1,900	133.00	133.40	500	17,900.00	2,150	26.65	26.95	100	-6.30	26.65	19.14	17,69,467	2,564	76,931	
	16,372	7,407	7,31,088	14.20	96.95	-54.05	50	95.50	97.00	1,250	17,950.00	250	38.25	40.25	200	-5.45	38.25	18.20	10,47,961	-1,118	29,981	
	79,656	23,408	21,74,871	14.42	66.00	-51.80	200	65.05	66.00	6,700	18,000.00	50	59.00	59.30	50	-1.70	58.50	18.35	16,31,212	-1,052	63,588	
	36,640	1,270	10,11,074	14.46	41.65	-47.55	3,450	41.00	41.65	50	18,050.00	100	83.25	84.90	50	2.20	82.60	18.00	4,17,856	-5,269	11,083	
	1,03,374	11,587	17,39,052	14.57	24.55	-40.05	100	24.25	24.55	50	18,100.00	50	116.75	117.90	500	12.60	118.00	19.24	3,92,532	-9,411	17,053	
	52,110	14,525	8,67,577	14.62	13.30	-30.55	50	13.25	13.50	1,150	18,150.00	50	155.45	156.30	50	22.90	156.90	20.29	53,725	140	3,812	
	1,25,965	11,666	15,28,823	14.69	6.65	-21.70	500	6.65	6.80	850	18,200.00	100	198.85	200.00	150	29.10	198.45	21.16	84,962	-2,825	5,071	
	52,363	17,713	6,58,412	15.18	3.50	-14.45	50	3.55	3.75	800	18,250.00	50	244.95	247.00	250	38.85	246.05	23.68	8,187	-224	2,426	
	93,123	6,713	9,06,681	16.80	2.65	-8.35	1,550	2.65	2.70	1,150	18,300.00	50	294.80	297.00	200	41.70	295.00	26.52	16,348	-1,559	2,377	
	30,319	-18,933	3,69,197	17.95	1.80	-4.85	1,350	1.85	2.00	50	18,350.00	100	342.60	345.95	100	54.70	346.30	30.37	731	-54	838	
	75,038	-6,256	6,57,196	20.19	1.80	-2.70	300	1.80	1.85	1,500	18,400.00	50	393.70	395.60	50	56.25	395.20	32.92	1,518	-302	699	
	21,834	-10,126	1,87,939	21.76	1.50	-1.90	1,500	1.50	1.55	1,450	18,450.00	100	442.20	447.10	100	45.00	435.10	29.05	56	6	109	

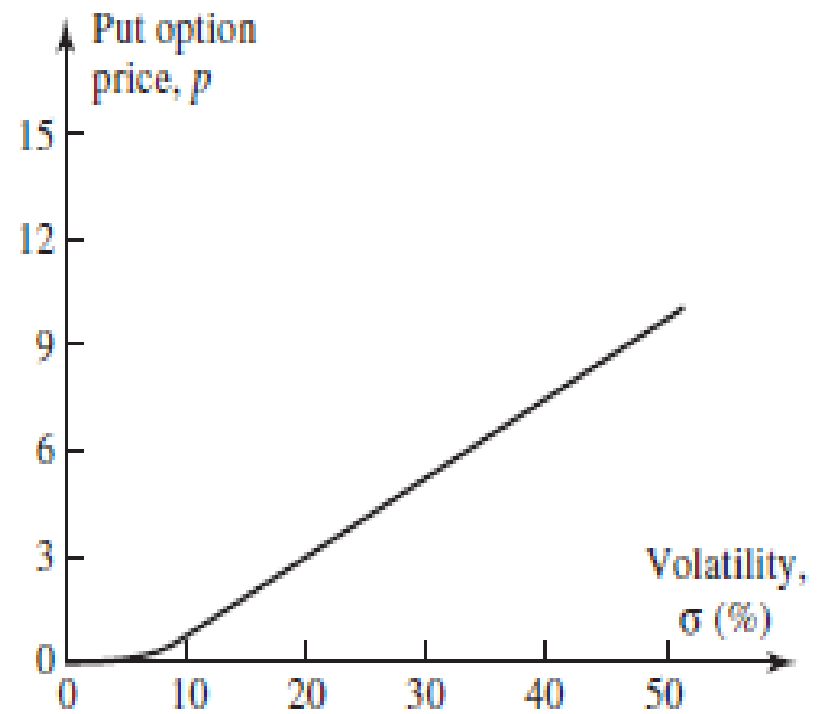
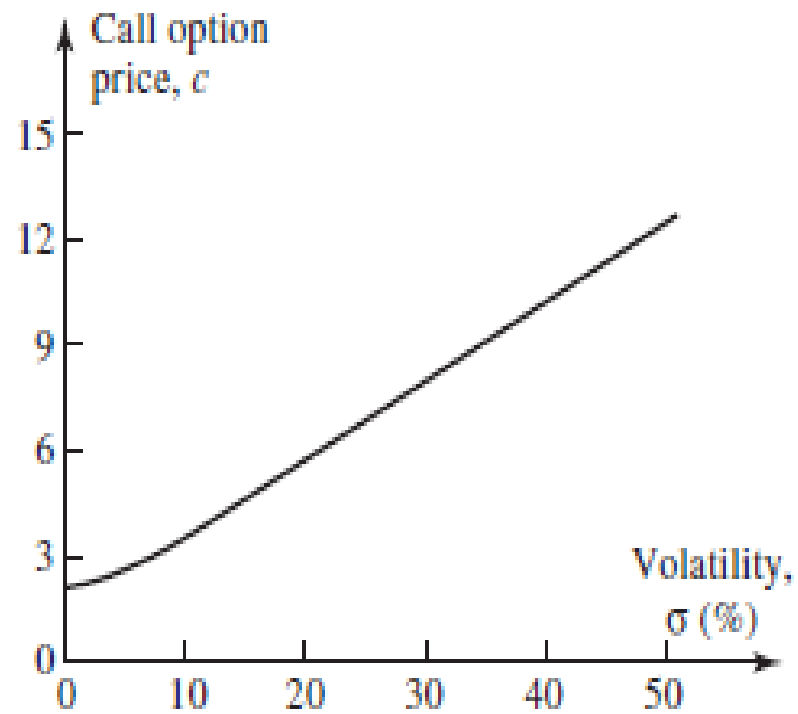
Time to expiration(American)

- While strike price has an impact on intrinsic value of the option, the time to expiration has a time value of the option
- The greater the time to expiration, the higher will be the probability of the option moving in the favour of the buyer. Hence, the higher will be the time value of an option subsequently, the higher will be the option price.



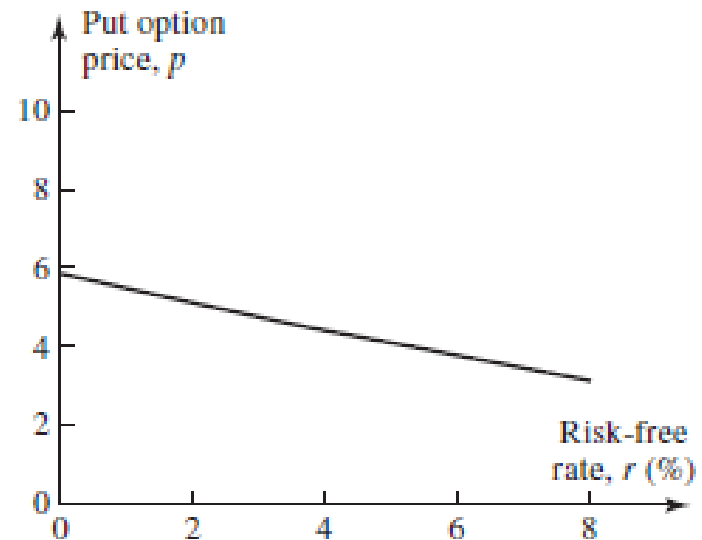
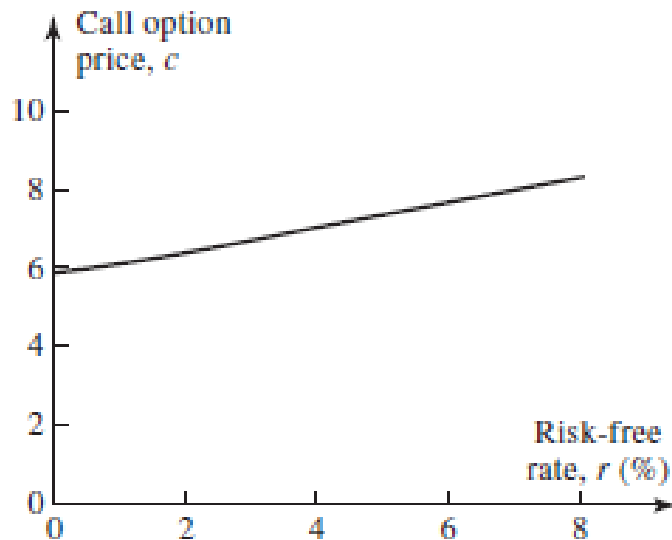
Volatility

- As volatility increases, the chance that the stock will do very well or very poorly increases.
- For the owner of a stock, these two outcomes tend to offset each other. However, this is not so for the owner of a call or put. The owner of a call benefits from price increases but has limited downside risk in the event of price decreases because the most the owner can lose is the price of the option.
- Similarly, the owner of a put benefits from price decreases, but has limited downside risk in the event of price increases. The values of both calls and puts therefore increase as volatility increases



Interest Rate

- As interest rates in the economy increase, the expected return required by investors from the stock tends to increase. In addition, the present value of any future cash flow received by the holder of the option decreases. The combined impact of these two effects is to increase the value of call options and decrease the value of put options.



Dividend

- Dividends have the effect of reducing the stock price on the ex-dividend date. This is bad news for the value of call options and good news for the value of put options.
- The value of the option is negatively related to the size of the dividend if the option is a call and positively related to the size of the dividend if the option is a put.

Upper bound of an option

An American or European call option gives the holder the right to buy one share of a stock for a certain price. No matter what happens, the option can never be worth more than the stock.

$$c \leq S$$

$$C \leq S$$

For a put option:

$$p \leq Ke^{-rT}$$

$$P \leq K$$

Lower Bound for European Call Option Prices; No Dividends

$$c \geq \max(S_0 - Ke^{-rT}, 0)$$

Arbitrage

- Suppose that

$$c = 3$$

$$S_0 = 20$$

$$T = 1$$

$$r = 10\%$$

$$K = 18$$

$$D = 0$$

- What are the arbitrage possibilities?

Lower Bound for European Put Prices; No Dividends

$$p \geq \max(Ke^{-rT} - S_0, 0)$$

Put-Call Parity: No Dividends

- Consider the following 2 portfolios:
 - Portfolio A: European call on a stock + zero-coupon bond that pays K at time T
 - Portfolio C: European put on the stock + the stock

Values of Portfolios

		$S_T > K$	$S_T < K$
Portfolio A	Call option	$S_T - K$	0
	Zero-coupon bond	K	K
	Total	$\bar{S}_T$	K
Portfolio C	Put Option	0	$K - S_T$
	Share	S_T	S_T
	Total	S_T	K

The Put-Call Parity Result

- Both are worth $\max(S_T, K)$ at the maturity of the options
- They must therefore be worth the same today. This means that

$$c + Ke^{-rT} = p + S_0$$

Arbitrage Opportunities

- Suppose that

$$c = 3$$

$$S_0 = 31$$

$$T = 0.25$$

$$r = 10\%$$

$$K = 30$$

$$D = 0$$

- What are the arbitrage possibilities when

$$p = 2.25 ?$$

$$p = 1 ?$$

Three-month put price = \$2.25

Action now:

Buy call for \$3

Short put to realize \$2.25

Short the stock to realize \$31

Invest \$30.25 for 3 months

Action in 3 months if $S_T > 30$:

Receive \$31.02 from investment

Exercise call to buy stock for \$30

Net profit = \$1.02

Action in 3 months if $S_T < 30$:

Receive \$31.02 from investment

Put exercised: buy stock for \$30

Net profit = \$1.02

Three-month put price = \$1

Action now:

Borrow \$29 for 3 months

Short call to realize \$3

Buy put for \$1

Buy the stock for \$31

Action in 3 months if $S_T > 30$:

Call exercised: sell stock for \$30

Use \$29.73 to repay loan

Net profit = \$0.27

Action in 3 months if $S_T < 30$:

Exercise put to sell stock for \$30

Use \$29.73 to repay loan

Net profit = \$0.27

Early Exercise

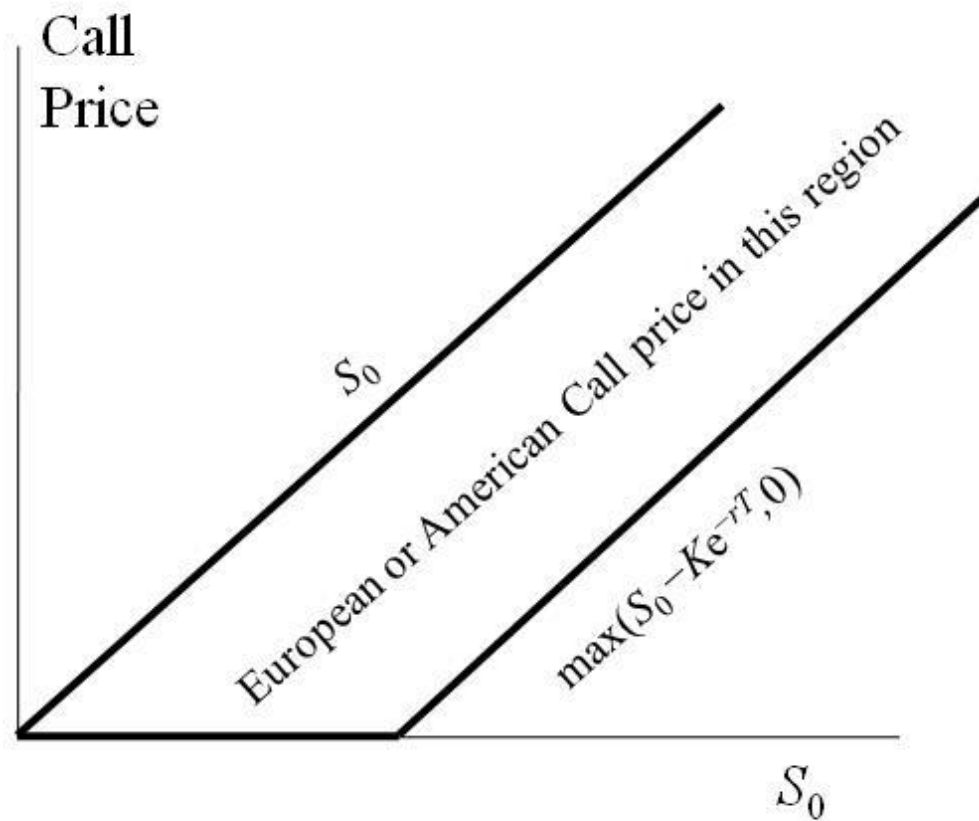
- Usually there is some chance that an American option will be exercised early
- An exception is an American call on a non-dividend paying stock
- This should not be exercised early



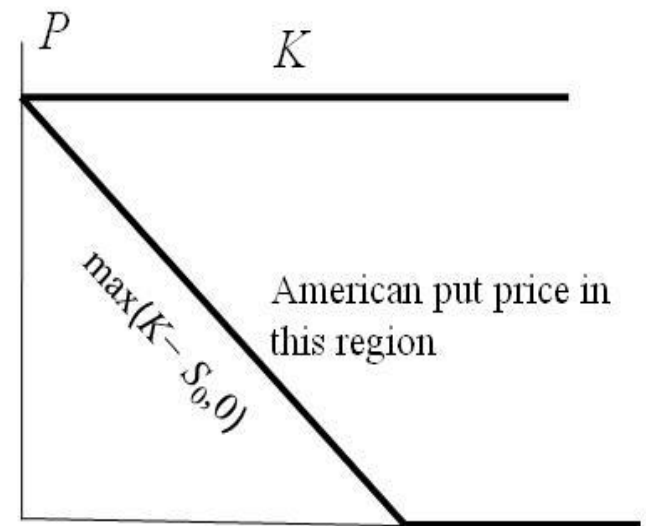
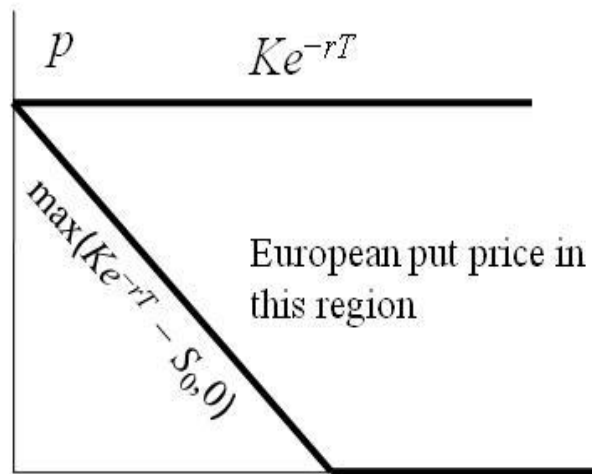
Reasons For Not Exercising a Call Early (No Dividends)

- No income is sacrificed
- You delay paying the strike price
- Holding the call provides insurance against stock price falling below strike price

Bounds for European or American Call Options (No Dividends)



Bounds for European and American Put Options (No Dividends)



The Impact of Dividends on Lower Bounds to Option Prices

$$c \geq S_0 - D - Ke^{-rT}$$

$$p \geq D + Ke^{-rT} - S_0$$

Extensions of Put-Call Parity

- American options; $D = 0$

$$S_0 - K < C - P < S_0 - Ke^{-rT}$$

- European options; $D > 0$

$$c + D + Ke^{-rT} = p + S_0$$

- American options; $D > 0$

$$S_0 - D - K < C - P < S_0 - Ke^{-rT}$$

Question- I

The price of a non-dividend paying stock is \$19 and the price of a three-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum with continuous compounding. What is the price of a three-month European put option with a strike price of \$20?

Question-2

A one-month European put option on a non-dividend-paying stock is currently selling for \$2.50. The stock price is \$47, the strike price is \$50, and the risk-free interest rate is 6% per annum with continuous compounding. What opportunities are there for an arbitrageur?

Question-3

The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. Risk-free interest rates (all maturities) are 10% per annum with continuous compounding. What is the price of a European put option that expires in six months and has a strike price of \$30?

Question-4

A European call option and put option on a stock both have a strike price of \$20 and an expiration date in three months. Both sell for \$3. The risk-free interest rate is 10% per annum, the current stock price is \$19, and a \$1 dividend is expected in one month. Identify the arbitrage opportunity open to a trader.